

GIORGIO MORRA

+1 (626) 563 8531 • Pasadena, California • giorgiomorra1@outlook.com

INTRODUCTION

Mechanical Engineering graduate student from Politecnico di Milano specializing in Mechatronics and Robotics, currently conducting Master's Thesis research at NASA JPL. I aim to contribute to innovative solutions for the control and navigation of Autonomous Systems.

EDUCATION

M.Sc. in Mechanical Engineering (Mechatronics and Robotics) 2023 – Ongoing

Politecnico di Milano, Italy

International Mobility Extra-EU: Pontificia Universidad Javeriana, Colombia

B.Sc. in Mechanical Engineering 2020 – 2023

Politecnico di Milano, Italy

Erasmus+ Program: Universidad Pontificia Comillas, Spain

EXPERIENCE

Robotics Navigation Intern Dec 2025 – Ongoing

NASA Jet Propulsion Laboratory – Robotic Mobility and Manipulation Section Pasadena, CA

Contributed to the **Venus Variable Altitude Aerobot** project, focusing on localization strategies for the Venusian atmosphere.

- Modeled the balloon's circumnavigation dynamic and interplanetary (Venus-Earth-Sun) communication;
- Simulated onboard measurements for sun sensor, altimeter, barometer and IMU;
- Designed the state prediction model and selected optimal filtering techniques;
- Performed sensitivity analyses to assess state estimation accuracy under varying sensor availability and noise levels;
- Developed a thermal FEM model to simulate the thermodynamics of the Zero Pressure (ZP) balloon.

UNIVERSITY PROJECTS

- Path Planning, Optimal Control and State Estimation;
- Collaborative Elbow Exoskeleton for Stroke Rehabilitation;
- SCARA Robotic Arm Mechanical Design, Simulation & Motion Planning;

SKILLS

Prog. Languages: Matlab, Python, C++, ROS2

Simulation Tools: Simulink, Simscape

CAD Software: Inventor, SolidWorks, Solid Edge, CATIA V5

Various: L^AT_EX, Microsoft Office

LANGUAGES

Italian (Native) — English (C1) — Spanish (C1)